

CSS - Cascading Style Sheet

- + it is a lng that is used to describe the style of a document.
- + (property: value) pair

Basic Syntax:

```
selector{  
    property: value;  
}
```

Ways of Including Style:

- + inline CSS: where we write Style inside HTML tags/elements as style attribute
`<h1 style="color: blue; text-align: center;">Welcome to Level 1 CSS</h1>`
`<tag style=" property1: value1; property2: value2; ....." > Content </tag>`

- + internal CSS: where we write Style inside `<style> </style>` tag we can include style tag inside head or inside body

```
<style>  
h1 {  
    font-family: Arial, sans-serif;  
    color: blue;  
    text-align: center;  
}  
</style>
```

- + external CSS: Writing CSS in separate file(style.css) and linking with HTML file(index.html)

```
selector{  
    property1: value1;  
    property2: value2;  
}
```

Note: inline Style got higher priority.

- + In external CSS if we styled a element with same property multiple time last property: value will apply

-
- + Color Property: used to set the color of four ground.

```
color: red;
```

- + background color: used to set the color of background.

```
background-color: red;
```

Color Systems

+ RGB (Red, Green, Blue) (0-255, 0-255, 0-255)

color: rgb(255,0,0); =>red

color: rgb(0,255,0); =>green

+ Hex(Hexadecimal) (#Red Green Blue)

(#00-ff 00-ff 00-ff)

color:#ff0000; =>red

color:#00ff00; =>green

+ Alpha Channel (RGBA): opacity (0 to 1)

color: RGBA(255,0,0,0.2);

color: RGBA(0,255,0,0.4);

Selectors

+ universal selector: *{ property: value; }

```
*{  
  margin: 0;  
  padding: 0;  
}
```

+ element selector: element{ property: value; }

```
h1{  
  color: red;  
}
```

+ id selector: #id{ property: value; }

```
#id3{  
  color: yellow;  
}
```

+ class selector: .class{ property: value; }

```
.myclass{  
  color: purple;  
}
```

+ group selector: selector1,selector2, ...{ property: value; }

```
#id1,#id2,#id3{  
  color: yellow;  
}
```

Text property:

- + text-align: left/right/center
- + text-decoration: underline/line-through/overline/none
- + font-weight: normal/bold/bolder or 100-900 (400=bold)
- + font-family: (font style) Times new Roman,arial ..etc
- + font-size: px / em / rem
- + line-height: (line spacing)
- + text-transform: uppercase/lowercase/capitalize/none
- + text-shadow: hr vr br color; (hr- horizontal ratio, vr- vertical ratio, br- blur ratio)

Units in CSS

- + Absolute units: pixel(px) 1 inch = 96px

Relative unites in CSS:

+ Percentage(%): It is often used to define a size as relative to an element's parent object

ex: width:33%;
margin-left:50%;

+ em : font size of the parent, in the case of typographical properties like front-size

and font size of the element itself, in case of other properties like width

+ rem(Root em):Font size of the root element (by default font size of body content is 16px)

+ vh: relative to 1% of viewport height

+ vw: relative to 1% of viewport width

BOX MODEL

+ Height: by default, it sets the content area height of the element

```
div{  
  height: 50px;  
  background-color: rgb(240, 240, 0);  
}
```

+ Width: by default, it sets the content area Width of the element

```
div{
  Width: 50px;
  background-color: rgb(240, 240, 0);
}
```

+ Border: used to set an element's border.

```
div{
  border-width: 2px;
  border-style: solid;
  border-color: black;
}
border: 3px solid black; // shorthand
border-radius: 10px/50%
```

+ Padding: space b/w content and border

```
padding-left:10px;
padding-top:20px;
padding-right:10px;
padding-bottom:20px;

padding:50px; //shorthand
padding: top right bottom left; -> clockwise padding: 10px 20px 10px 20px
```

+ Margin: space out side the border and space b/w 2 elemnts

```
Margin-left:10px;
Margin-top:20px;
Margin-right:10px;
Margin-bottom:20px;

Margin:50px; //shorthand
Margin: top right bottom left; -> clockwise Margin: 10px 20px 10px 20px
```

+ box-sizing: box size will be fixed it will not increase when we give padding

```
box-sizing: border-box;
```

+ overflow: text content overflow from container

```
overflow: auto/hidden/scroll;
```

+ Display Property:

display: inline/block/inline-block/none/flex

+ inline: Takes only the space required by the element (size of the content).(no margin/padding)

+ block: takes full space available in width.

+ inline-block: Similar to inline but we can set margin and padding

+ none: to remove element from document flow. (space for the element is not reserved)

+ visibility Property:

visibility: hidden

Note: when visibility is to hidden, space for the element is reserved but if display is set to none, space for the element is not reserved

+ Position: the position CSS property sets how an element is positioned in a document.

position: static/relative/absolute/fixed

+ static: by default position (the top, right, left, bottom and z-index properties have no effect)

+ relative: element is relative to itself.(the top, right, left, bottom and z-index properties will work)

+ absolute: position relative to its closest positioned(non static) ancestor

+ fixed: position relative to browser (it will not scroll)

+ sticky: positioned based on user's scroll position

+ background Image

background-image: url(download.jpeg);

background-repeat: no-repeat;

background-size: cover/contain/auto;

+ Flex Box: It is a one-dimensional layout method for arranging items in rows or columns.

+ The Flex Model:

- flex container
- flex item
- main axis
- cross axis

+ Flexbox Direction: It sets how flex items are placed in the flex container, along which axis and direction.

- flex-direction : row; (default)
- flex-direction : row-reverse;
- flex-direction : column;
- flex-direction : column-reverse;

+ Flex Properties (for Flex Container)

- justify-content : alignment along the main axis.
- flex-start / flex-end / centre / space-evenly /
- flex-wrap : nowrap / wrap / wrap-reverse
- align-items : alignment along the cross axis.
- align-content : alignment of space between & around the content along cross-axis

+ Flex Properties (for Flex Item)

- align-self : alignment of individual along the cross axis.
- flex-grow : how much a flex item will grow relative to the rest of the flex items if space is available
- flex-shrink : how much a flex item will shrink relative to the rest of the flex items if space is available

+ Media Queries

```
@media(width: _ px){  
    style  
}
```

```
@media(min-width: _ px)(max-width: _ px){  
    div{  
        style  
    }  
}
```

+ pseudo class: it use to style the state of the element

```
+ div: hover{  //(mouse hover)
    style
}
+ div: active{  //(mouse click)
    style
}
```

+ Transitions: transitions enables you to define the transition between two states of an element

transition-property: property you want to transition (font-size, width etc)
transition-duration: 2ms/4ms...
transition-timing-function: ease-in/ease-out/linear/steps..
transition-delay: 2s/4ms..
transition shorthand:
 transition : property | duration | timing-function | delay

CSS Transform: Used to apply 2D & 3D transformations to an element

+ rotate

```
transform: rotate(45deg)
```

+ scale

```
transform: scale(2);
transform: scale(0.5);
transform: scale(1, 2);
```

x & y axis separately:
transform: scaleX(0.5);
transform: scaleY(0.5);

+ translate

```
transform: translate(20px);
transform: translateX(20px);
transform: translateY(20px);
transform: translate(20px, 50px);
```

+ skew

```
transform: skew (30deg);
```

+ Animation: To animate CSS elements

```
@keyframe myName {  
  from { font-size : 20px; }  
  to { font-size : 40px; }  
}
```

+ Animation Properties:

```
animation-name:  
animation-duration:  
animation-timing-function:  
animation-delay:  
animation-iteration-count:  
animation-direction:
```

+ Animation Shorthand

```
          animation-name | duration | timing-function | delay | iteration |  
direction  
          animation : myName 2s linear 3s infinite normal
```

+ % in Animation

```
@keyframe myName {  
  0% { font-size : 20px; }  
  50% { font-size : 30px; }  
  100% { font-size : 40px; }  
}
```